CSTANet: Cross-Frame Spatio-Temporal Aggregation Network for Moving Small-Object Detection in Satellite Videos

Xinrui Qi, Siqian He, Wuzhuo Yang, Yuzeng Chen, Xin Su, *Member, IEEE*, and Qiangqiang Yuan, *Member, IEEE*

***Abstract*—With the rapid advancement of satellite video technology, detecting small moving objects in satellite videos has shown substantial application potential across domains such as military surveillance, traffic monitoring, and disaster management. However, achieving robust detection remains challenging due to three key factors: 1) limited spatial resolution; 2) complex and dynamic backgrounds; and 3) low object-background contrast. To address these issues, we propose a cross-frame spatio-temporal aggregation network (CSTANet), which deeply exploits and effectively fuses spatio-temporal features to enhance detection accuracy and robustness. Extensive experiments conducted on satellite video moving object detection (MOD) datasets demonstrate that CSTANet outperforms state-of-the-art methods in multiple metrics, including recall, precision, and $F1$-score. This study provides an effective solution for detecting small moving objects in satellite videos and is expected to advance the development of downstream remote sensing surveillance missions.**

***Index Terms*—Cross-frame aggregation, moving object detection (MOD), satellite video, spatio-temporal feature mining.**

I. Introduction

VIDEO satellites provide continuous Earth observation with wide coverage, high temporal resolution, and rich spatio-temporal context [1], supporting applications such as military surveillance, wildfire monitoring, and traffic analysis. Among these applications, moving object detection (MOD) in satellite videos plays a fundamental role. MOD aims to localize and identify objects exhibiting spatial semantic coherence and sustained temporal motion. Robust detection of diverse objects in satellite videos is crucial for effective surveillance and cross-domain adaptability. However, satellite video imaging presents several inherent challenges.

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1) *Limited Spatial Resolution:* Vehicles in satellite videos typically occupy only 8 × 8 to 12 × 12 pixels [2], significantly constraining discriminative feature extraction. The small object size leads to a deficiency in appearance-based attributes such as texture and geometric cues, complicating reliable identification.
2) *Complex and Dynamic Backgrounds:* Satellite video frames contain numerous moving objects embedded in complex non-stationary backgrounds. In addition, the motion of satellite platforms induces imaging artifacts, increasing false alarms and disrupting detection accuracy.
3) *Low Object-Background Contrast:* The interaction of complex imaging backgrounds and sensor noise often obscures target objects, diminishing contrast and intensifying detection challenges.

To visually highlight the challenges associated with detecting small moving vehicles in large-scale satellite videos, Fig. 1 illustrates some examples from the Jilin-1 video satellite objects (VISOs) dataset [3].

Advancements in satellite video processing have significantly improved small-object detection capabilities. Current multi-object detection methods for satellite videos can be broadly classified into traditional and deep-learning-based approaches. Traditional methods include background subtraction [4], [5], [6], [7], frame difference [8], [9], [10], and optical flow [11], [12], [13], which detect motion using foreground models, inter-frame intensity changes, or pixel-wise motion. However, their handcrafted assumptions make them vulnerable to dynamic backgrounds, platform-induced pseudo-motion, noise, illumination changes, and high computational cost, limiting robust detection of weak or slow-moving small objects.

Deep learning methods provide more adaptive spatial and semantic representations. Existing approaches mainly include single-frame and multi-frame networks. Single-frame methods enhance appearance features through multi-scale fusion, super-resolution, contextual modeling, and pyramid aggregation [14], [15], [16], [17]. However, they ignore temporal motion and may confuse weakly moving objects with static structures or noise in complex satellite-video scenes.

Multi-frame networks exploit consecutive frames through optical-flow or frame-difference [18], [19], model driven meth-

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Fig. 1. Examples of moving vehicle detection in the VISO dataset, arranged from top to bottom: (a) video 002, (b) video 003, (c) video 008, and (d) video 010.

ods [20], [21], 3-D convolution-based methods [22], [23], recurrent or memory-based methods [24], [25], [26], [27], and attention or Transformer enhanced methods [28], [29]. These strategies remain limited by noisy motion priors, restrictive assumptions, high computation or motion smoothing, accumulated temporal errors, and insufficient handling of tiny-object motion, platform-induced displacement, background dominance, and spatio-temporal misalignment.

Despite this progress, existing methods often enhance either spatial appearance or temporal context without adequately handling the coupling between weak object motion and satellite-platform motion. Noisy motion priors and direct frame stacking may dilute sparse motion cues, while concatenation, channel-wise addition, or late-stage weighting can propagate misaligned temporal artifacts. Therefore, an effective detector for satellite videos should address two key issues: 1) it should extract temporal cues that are sensitive to tiny moving objects, rather than merely treating frame differences as handcrafted foreground maps, and 2) it should align temporal motion features with current frame spatial semantics before feature integration.

To address these limitations, we propose a cross-frame spatio-temporal aggregation network (CSTANet) for moving small object detection in satellite videos, as illustrated in Fig. 2. CSTANet follows a dual-branch design, in which the spatial feature extraction (SFE) branch captures appearance, texture, and contextual information, while the temporal feature extraction (TFE) branch explores cross-frame motion cues. CSTANet employs a cross-frame temporal difference learning (CTDL) module that embeds cross-frame difference information into an end-to-end learnable temporal branch. By aggregating bidirectional cross-frame differences and progressively compensating the current-frame representation, CTDL transforms weak inter-frame variations into local motion residuals while reducing background fluctuations and platform-induced pseudo-motion.

Furthermore, to avoid treating temporal cues as simply appended auxiliary features, CSTANet designs a motion compensation unit (MCU) for spatio temporal feature fusion. MCU models the residual discrepancy between temporal features and current-frame spatial features and compensates for feature misalignment before aggregation. The compensated features are then fused through a hierarchical multi-scale aggregation strategy to integrate motion cues with spatial appearance features.

The main contributions of this work are as follows.

1) The CSTANet for moving small object detection in satellite videos. It addresses the limitations of existing deep learning methods by jointly enhancing weak temporal motion representation and alignment aware spatio temporal feature fusion.
2) We design a CTDL module that aggregates bidirectional inter-frame differences and progressively compensates the current-frame representation to preserve subtle motion cues of small moving objects.
3) We propose an MCU module to perform residual-based compensation between temporal and spatial features before feature fusion, reducing feature misalignment caused by satellite platform motion and parallax.
4) Extensive experiments are conducted on SVMOD [23] and SatMTB-MOT [30]. SVMOD [23] systematically re annotates weak featured and small-scale objects based on the VISO dataset [3]. Experimental results demonstrate that CSTANet outperforms state-of-the-art methods in both qualitative and quantitative evaluations.

This article is structured as follows: Section II reviews related work. Section III details the proposed CSTANet network. Section IV describes the dataset, experimental setup, and evaluation metrics, followed by an analysis of the results. Section V concludes the study and summarizes key findings.

II. Related Work

A. Traditional Object Detection Methods

Common traditional methods for object detection in videos mainly include background subtraction [4], [5], [6], [7], frame difference [8], [9], [10], and optical flow [11], [12], [13]. Representative approaches include GMMv2 [31] and E-LSD [4] for background modeling, FFD [32] and E-GMMTFD [33] for interframe differences, and edge-constrained optical flow [13] and improved ViBe-based detection [34]. E-3DTV [35] further combines 3-D total variation with Gaussian priors in a low-rank and sparse decomposition framework. However, background subtraction [4], [5], [6], [7] is vulnerable to

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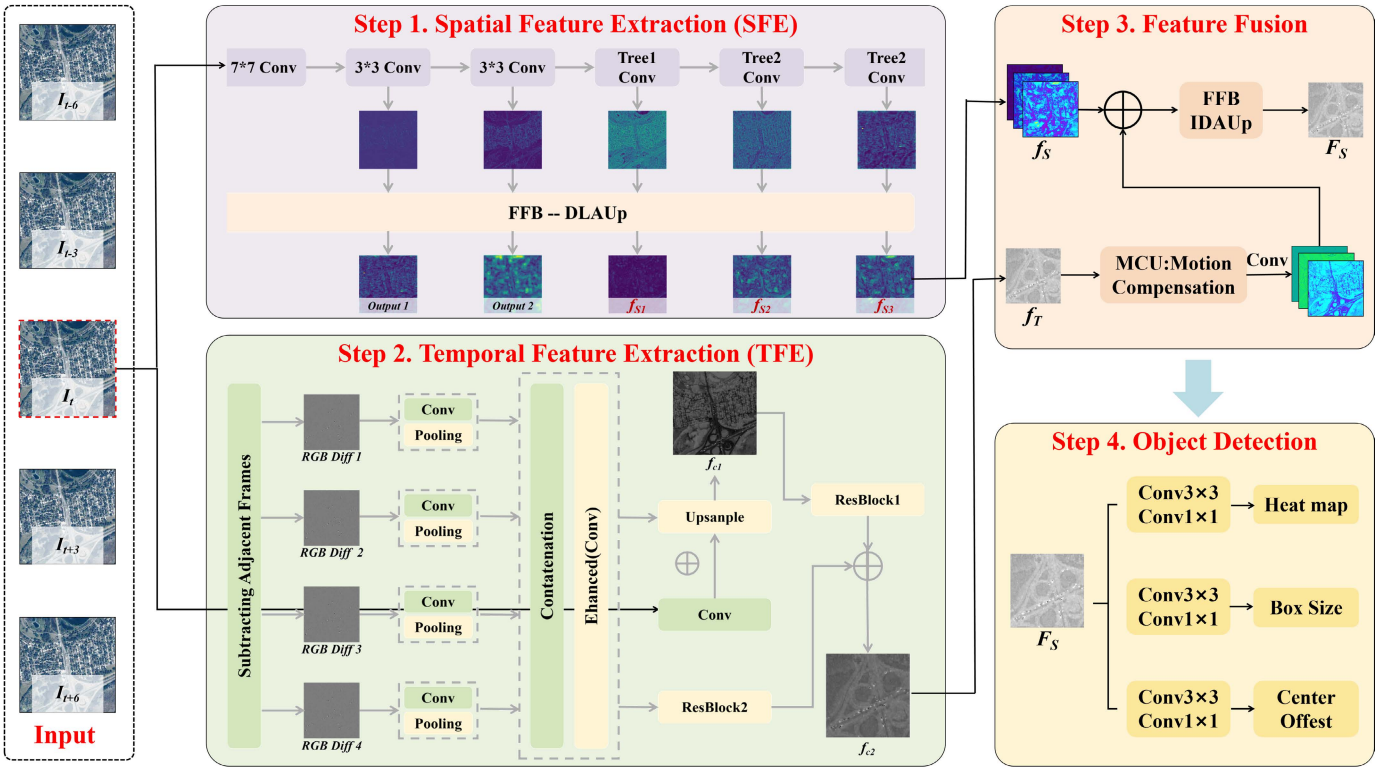


Fig. 2. Overall architecture of the proposed CSTANet. It comprises three main modules: the SFE module, the TFE module, and the feature fusion module.

dynamic backgrounds and illumination changes. Frame difference methods [8], [9], [10] often misses slow or extremely small targets. Optical flow methods [11], [12], [13] is computationally expensive and noise-sensitive. These limitations hinder robust and efficient detection in large-scale, high-resolution satellite videos.

B. Object Detection Methods Based on Deep Learning

Deep learning improves feature representation and adaptability for moving small-object detection. Existing methods can be mainly divided into 1) single-frame detection networks; 2) multi-frame detection networks; and 3) Transformer-based methods.

1) Single-Frame Detection Networks: Single-frame convolutional neural network (CNN) detectors improve spatial representation through feature fusion, architecture design, and data augmentation. A feature-fusion and scaling detector [16] incorporates spatial context, while EFPN [14] uses high-resolution pyramids, texture transfer, and balanced loss. CenterNet [36] performs keypoint-based localization, and YOLOX [37] adopts dynamic label assignment and a decoupled head. GAN-based augmentation [38] and AugTarget [39] further enrich training samples. However, single-frame methods rely only on appearance, texture, and contextual features while ignoring temporal motion. Consequently, tiny low-contrast objects can be confused with background structures or noise in complex satellite-video scenes.

2) Multi-Frame Object Detection Networks: Multi-frame networks exploit temporal information from consecutive satellite-video frames to enhance motion representation and detection performance. According to different temporal modeling strategies, existing multi-frame detection networks can be divided into optical flow or frame difference driven methods [18], [19], model driven methods [20], [21], 3D convolution based methods [22], [23], recurrent or memory based methods [24], [25], [26], [27], and attention or Transformer enhanced methods [28], [29].

Optical-flow and frame-difference methods directly extract pixel-level motion cues. SFOD-SAT [18] extracts motion and appearance features from optical flow and video frames and fuses them through an attention module for semi-supervised detection. LSCNet [19] highlights moving pixels using multi-frame differences and aggregates consecutive-frame features for real-time detection. Model-driven methods formulate detection as an optimization problem. DMODNet [20] unfolds ADMM-based low-rank and sparse decomposition into a learnable network, while S-SPL [21] combines scene-level priors with collaborative self-paced learning for unsupervised detection. 3D-convolution-based methods model spatial and temporal information jointly. DSFNet [22] adopts a dual-stream structure to extract spatial semantics and temporal features, whereas Hieum [23] represents multi-frame foreground pixels as sparse spatio-temporal point clouds and applies sparse convolution for long-term modeling. Recurrent methods capture temporal dependencies through hidden states. SDANet [24] uses bidirectional Conv-RNN for temporal feature extraction and alignment. SPAX [25] employs ConvGRU in a hierarchical spatio-temporal fusion framework. MP^{2}Net [26] combines ConvGRU with mask propagation to compensate for

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inter-frame displacement. MICPL [27] integrates multi-layer ConvLSTM with a motion-visual adapter. Attention-based methods emphasize informative spatial and temporal features. MLSTA-Net [28] uses multi-level spatio-temporal attention to model cross-frame consistency. GHOST-R [29] combines deformable attention, temporal semantic reasoning, and graph neural networks for high-order relation modeling. MICPL [27] further introduces motion-vision relation attention, whereas DNTR and DConvTrans-LGA use Transformer structures to strengthen long-range dependency modeling and local–global feature representation.

Nevertheless, flow or difference methods [18], [19] are sensitive to pseudo-motion; model-driven methods [20], [21] depend on restrictive priors; 3D convolution methods [22], [23] increase computation and may smooth weak motion; recurrent methods [24], [25], [26], [27] accumulate temporal errors; and attention-based methods [28], [29] still inadequately address weak motion, platform-induced displacement, and spatio-temporal misalignment.

3) Transformer-Based Methods: Recent Transformer and sequence models have been explored in remote-sensing video analysis. TLSH-MOT [40] combines a vision Transformer, spatio-temporal memory, and IDLinker for drone-view MOT. MemTrack [41] uses prior knowledge and temporal memory for satellite tracking. SiamTITP [42] integrates temporal information and trajectory prediction. An end-to-end MOT framework for vehicles and anomalous states in remote sensing videos [43] combines detection Transformer (DETR), trajectory guidance, Mamba blocks, and sparse attention.

However, these studies mainly address trajectory association, single-object or infrared tracking, and general temporal aggregation. Satellite-video multi-object detection remains underexplored, particularly for extremely small targets, weak appearance features, platform-induced pseudo-motion, and spatio-temporal misalignment.

Based on the above analysis, CSTANet adopts two task-oriented components for moving small-object detection in satellite videos. The CTDL module incorporates bidirectional inter-frame differences into a learnable temporal branch to capture local motion cues and reduce the influence of background fluctuations and pseudo-motion. The MCU models the residual discrepancy between temporal and current-frame spatial features and performs feature-level compensation before aggregation, helping alleviate spatio-temporal misalignment. Together, CTDL and MCU provide a practical combination of temporal-difference modeling and feature-fusion strategies for complex satellite video scenes. The detailed architecture is presented in Section III.

III. Proposed Approach

In this section, we provide a detailed introduction of the proposed CSTANet, as shown in Fig. 2. It mainly includes three modules: the SFE module, the TFE module, and the feature fusion module. In this network, the current frame is first fed into the SFE module. The deep layer aggregation-34 (DLA-34) [17] is then employed to extract spatial features such as the appearance, texture, and contextual information of the objects. Next, in the TFE module, we design a CTDL network. Centered on the current frame, this network takes two frames forward and two frames backward, with a frame difference of three between each pair of frames, to explore the temporal cues of the objects. Finally, in the feature fusion module, we design an MCU. In particular, the temporal cues and spatial features are first input into the MCU to ensure the alignment of temporal and spatial features, and then the IDAUp module is used to complete the fusion of spatio-temporal features.

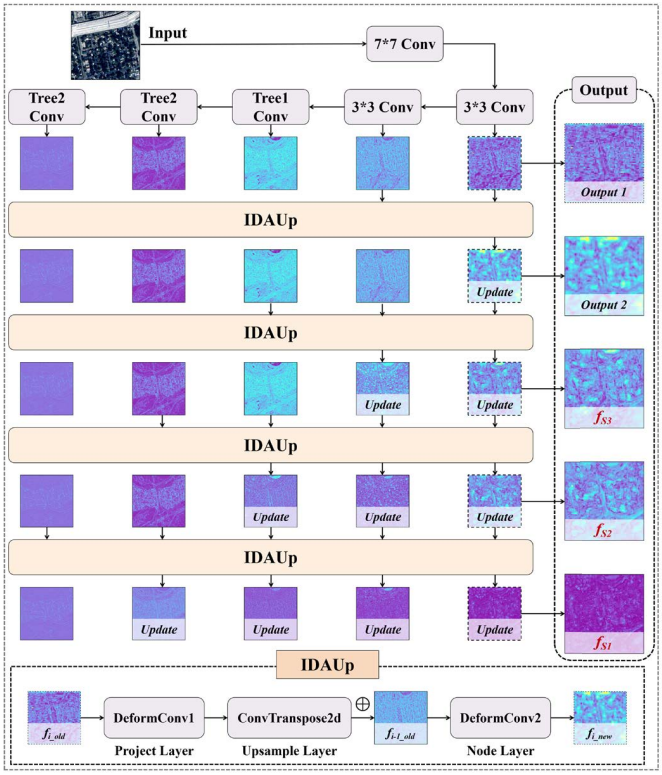


Fig. 3. Overall architectural diagram of the SFE module uses the DLA-34 backbone network to extract spatial features of the object. Then, the DLAUp network is employed to achieve multilevel feature fusion, ultimately producing a three-layer enhanced spatial feature of the moving object.

A. Spatial Feature Extraction Module

In the SFE module, we refer to the work in [17] and use the DLA-34 backbone network to extract the spatial features of the objects. After obtaining the multi-scale and multi-level features of the objects, we employ the deep layer aggregation up (DLAUp) network to complete the fusion of these multi-level features. Finally, we obtain three layers of enhanced spatial features for moving objects. The overall structure of our SFE module is shown in Fig. 3.

The DLA-34 is a convolutional neural network based on DLA, structured into six hierarchical levels for systematic feature extraction and aggregation. The first level employs a 7×7 convolutional layer with batch normalization (BN) and ReLU activation to capture low-level features. The next two levels use multiple 3×3 convolutional layers for shallow feature extraction, with spatial downsampling via strided convolutions. These layers gradually downsample

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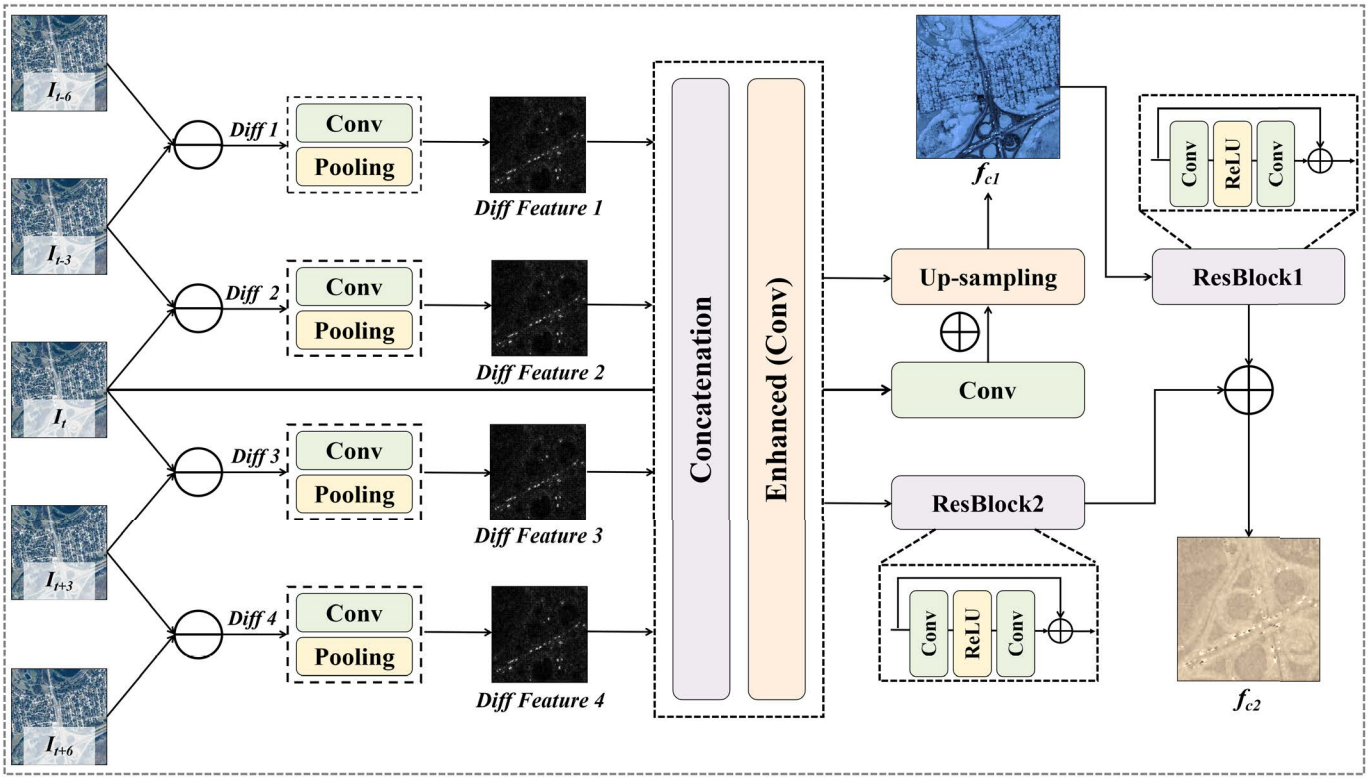


Fig. 4. Overall architectural diagram of the TFE module. We independently designed a CTDL network to deeply explore the temporal cues of the object.

feature maps through strided convolutions, reducing spatial dimensions while increasing feature richness. The final three levels feature tree-structured Tree modules. These modules adopt multi-branch architectures to aggregate features from different layers and resolutions, enabling the extraction of hierarchical, high-level semantic information. As the network progresses hierarchically, the feature map resolution decreases, while the number of channels increases exponentially, facilitating the capture of complex semantic patterns. In practice, DLA-34 successfully extracts five hierarchical spatial features of moving objects, demonstrating its effectiveness in multi-scale feature representation for object analysis tasks.

After extracting the five hierarchical spatial features, we apply the DLAUp method for multi-scale and multi-level feature fusion. DLAUp consists of four layers of IDAUp modules, with the number of modules per layer determined by the input feature map count. The network processes feature maps in ascending order of resolution. For each layer, the IDAUp module performs up-sampling and fusion through three steps: 1) *Channel Adjustment:* Projection layers recalibrate the number of channels in the feature maps; 2) *Resolution Matching:* Deconvolutional layers up-sample the feature maps to match the resolution of the upper-layer features; and 3) *Feature Fusion:* The current layer's features are point-wise added and fused with upper-layer features via a node layer, generating enhanced representations. This layer-wise processing integrates multiscale information, boosting feature discriminability, as follows:

$$f_{\text{output}} = \text{Node}\left(\text{Upsample}\left(\text{Project}\left(f_{\text{current}}\right)\right) \oplus f_{\text{before}}\right). \quad (1)$$

After feature fusion with DLAUp, five feature maps are generated. However, to mitigate potential semantic feature loss of small objects during down-sampling convolution, we select the first three layers of feature maps as the SFE module output. These enhanced maps retain both the precise location and detailed information of moving objects, along with contextual semantic information of small moving objects in satellite videos, providing rich feature support for subsequent detection tasks.

B. Temporal Feature Extraction Module

In the TFE module, we independently design a CTDL network to deeply explore the temporal cues of the object. This network centers on the current frame, extracting two frames from the forward and backward directions, with an inter-frame interval set to 3. First, red, green, and blue (RGB) difference maps of adjacent frames are calculated using differential operations. These difference maps are then converted into local motion representations, which are provided to the object frame through lateral connections. The overall structure of the TFE module is shown in Fig. 4.

Since moving objects in satellite videos usually occupy only a few pixels and exhibit low object-background contrast, their inter-frame variations are sparse and subtle. A single short-term difference may therefore fail to provide sufficiently discriminative motion responses. Meanwhile, complex dynamic backgrounds and satellite platform motion often introduce background fluctuations, pseudo-motion artifacts, and irregular pixel offsets, making direct frame-difference cues unreliable. To address these issues, CTDL aggregates

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and enhances bidirectional short-term differences rather than relying on an individual difference map. In particular, the network first computes temporal differences between adjacent frames and separately extracts the initial features of the current frame and the difference features. The difference features are then down-sampled, stacked, and enhanced by convolutional operations to obtain a more robust local motion representation. This design converts weak inter-frame variations into learnable motion residuals while reducing the influence of noisy background responses.

After obtaining the connection-enhanced difference features, we further introduce a progressive two-stage difference compensation strategy. The motivation is to decouple coarse motion enhancement from fine local refinement. In the first stage, the enhanced difference features are up-sampled to the current-frame resolution using bilinear interpolation and then combined with the initial features of the current frame through weighted fusion, generating the preliminary compensated features f_{c1}. The formula for the first stage of compensation is as follows:

$$f_{c1} = \alpha \cdot Up\left(f_{\text{diff-enhanced}}\right) + \beta \cdot f_{\text{initial}} \tag{2}$$

where f_{c1} represents the object frame features generated in the first stage. $f_{\text{diff-enhanced}}$ denotes the enhanced difference features. f_{initial} denotes the initial features of the object frame. α and β are weighting coefficients, and $Up(\cdot)$ represents the bilinear interpolation up-sampling operation. This stage introduces a coarse motion prior and promotes feature-level consistency between the current frame and its temporal neighbors, which helps enhance weak motion cues of tiny vehicles.

In the second stage of compensation, which is also called fine refinement, the preliminary compensated features f_{c1} are further enhanced through residual blocks. Meanwhile, the enhanced difference features are processed by residual blocks and up-sampling operations to generate refined compensation features. These two branches are fused again to produce the final temporal feature representation f_{c2}. The formula for the second stage of compensation is as follows:

$$f_{c2} = \alpha \cdot ResBlock1\left(f_{c1}\right) + \beta \cdot Up\left(ResBlock2\left(f_{\text{diff-enhanced}}\right)\right) \tag{3}$$

where f_{c1} represents the object frame features generated in the first stage and f_{c2} denotes the object frame features generated in the second stage. This stage focuses on fine-grained residual correction, which suppresses noisy difference responses caused by dynamic backgrounds and platform-induced pseudo-motion while preserving local details and boundary cues of small moving objects. Through this progressive compensation process, CTDL effectively transforms sparse cross-frame variations into discriminative temporal cues, serving as a reliable complement to spatial features.

Through the aforementioned process, the CTDL module effectively leverages temporal difference information from adjacent frames to incrementally optimize and enhance the features of the object frame. Ultimately, this network generates an object temporal feature map that encompasses extensive temporal cues of the object, serving as a complement to spatial features.

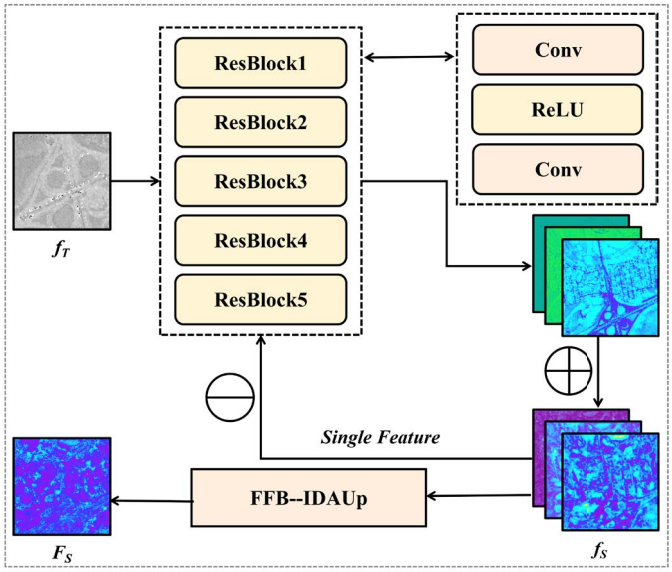


Fig. 5. Overall architecture diagram of the spatio-temporal feature fusion module. First, we designed an MCU, followed by employing IDAUp to complete the spatio-temporal feature fusion.

C. Spatio-Temporal Feature Fusion

After extracting spatial and temporal features of moving objects, spatio-temporal feature fusion is required. To achieve rational and effective integration, we design a novel spatio-temporal feature fusion strategy. First, an MCU is developed, which takes temporal cues from the TFE module and spatial features as inputs to ensure alignment and avoid spatio-temporal mismatches. Subsequently, the multiscale feature fusion module IDAUp—proposed in the DLA framework—is employed to complete spatio-temporal feature fusion. The overall architecture of our spatio-temporal feature fusion module is shown in Fig. 5.

Since temporal cues are extracted from the current frame and its adjacent frames, they may contain not only object-related motion information but also background fluctuations from neighboring frames. In satellite videos, temporal and spatial features are not naturally aligned. Limited spatial resolution makes the localization of small objects highly sensitive to slight pixel shifts, while complex dynamic backgrounds and satellite platform motion may cause parallax, pseudo-motion, and spatial offsets between frames. In addition, low object-background contrast makes temporal responses more likely to be mixed with background interference. Therefore, directly concatenating or adding temporal and spatial features implicitly assumes that they are spatially consistent, which may propagate misaligned temporal artifacts into the detection head. To overcome this limitation, we design an MCU to perform feature-level compensation before spatio-temporal aggregation.

The MCU module includes five residual blocks, each containing two convolutional layers that perform feature extraction operations. After the convolutional layers, normalization and activation functions can be applied. Input features are sequentially processed through the five residual blocks, with each

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block progressively enhancing and optimizing the features. Within the residual blocks, input features are combined with the output of the convolutional operations through skip connections, forming residual mappings. Regarding temporal and spatial features, we first calculate the residual features between the spatial feature f_S and the temporal feature f_T. These residual features are then fed to the MCU for enhancement and provided to f_T. In this way, temporal features are compelled to focus on valuable compensatory information from spatial features. The difference in compensation between f_S and f_T can be summarized by

$$f_T' = f_S + MCU\,(f_S - f_T) \tag{4}$$

where f_T' is the motion-compensated temporal feature.

After completing the motion compensation for temporal features, we use convolution operations to convert a single temporal feature map into three temporal feature maps, matching the dimensions of the three spatial feature maps. Subsequently, these spatio-temporal feature maps with corresponding dimensions are element-wise added, and the fused feature map is sent to the multi-scale feature fusion module, IDAUp, for multi-level feature fusion. Finally, the fused feature map is input to the detection head for completing the object detection task.

Through the aforementioned process, we ensure a close alignment of temporal and spatial characteristics to address potential issues caused by spatio-temporal misalignment. By utilizing the IDAUp module, we achieve a comprehensive integration of spatio-temporal features, enabling the interdependently beneficial enhancement of temporal and spatial attributes and thereby enhancing the detection accuracy of moving objects.

D. Anchor-Free Detector

Anchor-based detectors [44], [45], [46] have long dominated the field of object detection. However, the anchor mechanism introduces excessive manually designed parameters, complex computational procedures, and sample imbalance issues, which limit the further development of anchor-based detectors. Furthermore, the small size of moving objects in satellite videos, coupled with inherent imaging defects, further reduces the detection performance of anchor-based detectors. On the contrary, anchor-free detectors [47], [48], [49] directly predict the object position without predefined anchors, exhibiting higher generalization capabilities. This characteristic makes them particularly suitable for detecting small moving objects in satellite videos.

The proposed CSTANet is inspired by CenterNet and adopts an anchor-free detector. Similar to CenterNet, we represent the object as the center point of its bounding box, and the other attributes of the object (such as size and position) are directly regressed using the image features of the center point. During detection, we use a fully convolutional network to generate a heatmap, in which the peaks correspond to the center points of the objects. Subsequently, the image features of each peak are used to predict the height and width of the object bounding box. As shown in the diagram, the detection head of the proposed CSTANet consists of three parallel branches: the heatmap, box-size, and center-offset branches. In particular, each branch comprises a 3×3 convolutional layer, a ReLU activation function, and a 1×1 convolutional layer.

IV. Experiments

A. Experimental Sets

1) Datasets: Two satellite video datasets are used in this experiment, namely SVMOD [23] and SatMTB-MOT [30]. SVMOD is a reannotated version of the Jilin-1 VISO dataset [3], with a ground sampling distance of 0.92 m and a frame rate of 10 frames/s. It mainly contains moving vehicles smaller than 50 pixels. To address missing dark objects and inaccurate labels in VISO, Xiao et al. [23] increased the test annotations from 93 491 to 155 987 instances. Following their setting, 72 video subsets containing 27 434 images are used for training, and seven subsets containing 2255 images are used for testing. Training and test images have resolutions of 512×512 and 1024×1024, respectively. The differences between the seven test sets before and after re-annotation are presented in Fig. 6. It is evident that the reannotated test sets show a significant increase in the number of objects with more accurate annotations.

SatMTB-MOT [30] contains 224 Jilin-1 satellite videos, including 62 airplane, 70 ship, and 92 vehicle sequences. We convert its tracking annotations into horizontal bounding boxes for detection evaluation.

For consistency, all annotations used in our experiments are represented as two-dimensional horizontal bounding boxes in the format of (x, y, w, h), where (x, y) represents the center coordinates of the object, and w and h denote the width and height of the bounding box, respectively.

2) Evaluation Metrics: Precision (Pr), recall (Re), and $F1$-score are used for evaluation. Precision measures the proportion of correct detections among all predictions, recall measures the proportion of ground-truth objects detected, and $F1$ balances both metrics.

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \tag{7}$$

Here, true positive (TP), false positive (FP), and false negative (FN) denote correct detections, false alarms, and missed objects, respectively. A prediction is matched to the ground truth (GT) according to Intersection over Union (IoU). Because small deviations strongly affect the IoU of tiny satellite-video objects, we use an IoU threshold of 0.2 to avoid excessively penalizing otherwise valid detections.

3) Implementation Details: Experiments are conducted with PyTorch on two NVIDIA RTX 2080Ti GPUs. Each input contains five frames sampled at an interval of 3. The batch size and initial learning rate are set to 2 and 1.25×10^{-4}, respectively. Random mirroring and color jitter are used for augmentation. The model is trained for 55 epochs using Adam,

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Fig. 6. Seven test scenes: (a) video 002, (b) video 003, (c) video 005, (d) video 006, (e) video 008, (f) video 009, and (g) video 010. For better visualization, the regions marked by red dashed boxes in each scene are enlarged. For example, scene (h) enlarged version of scene (g). In each scene, red detection boxes represent newly added annotated objects, while green detection boxes indicate previously annotated objects.

with the learning rate reduced by a factor of 0.1 at epochs 30 and 45.

Following CenterNet [36], the total loss function is defined as a weighted combination of three components; $Loss_{total}$ is defined as follows:

$$Loss_{total} = \lambda L_{hm} + \eta L_{box} + \gamma L_{offset} \tag{8}$$

where L_{hm} denotes the heatmap loss, L_{box} represents the bounding box regression loss, and L_{offset} is the center offset loss. The three coefficients are empirically set to $\lambda = 1$, $\eta = 0.1$, and $\gamma = 1$. The definitions of the three individual loss functions are different from each other. First, L_{hm} is defined as follows:

$$L_{hm} = -\frac{1}{N}\sum_{x,y} \begin{cases} \left(1-\hat{Y}_{xy}\right)^{\alpha}\log\left(\hat{Y}_{xy}\right), & if\ Y_{xy}=1 \\ \left(1-Y_{xy}\right)^{\beta}\left(\hat{Y}_{xy}\right)^{\alpha}\log\left(1-\hat{Y}_{xy}\right), & if\ Y_{xy}\in[0,1) \end{cases} \tag{9}$$

where N is the number of training samples. Y_{xy} is the GT at location (x, y). $\hat{Y}_{xy}$ is the predicted value. $\alpha = 2$, $\beta = 4$ are hyperparameters. This formulation, similar to focal loss, alleviates the imbalance between positive and negative samples and emphasizes hard-to-detect objects. Second, L_{box} is consistent with a standard *L1 loss*

$$L_{box} = \frac{1}{N}\sum_{k=1}^{N}\left\|\hat{b}_k - b_k\right\|_1 \tag{10}$$

where $\hat{b}^k = (\hat{w}^k, \hat{h}^k)$ are the predicted width and height and $b^k = (w^k, h^k)$ are the ground-truth values. Finally, L_{offset} is also consistent with a standard *L1 loss*

$$L_{offset} = \frac{1}{N}\sum_{k=1}^{N}\|\hat{o}_k - o_k\|_1 \tag{11}$$

where $\hat{o}_k$ is the predicted offset of the object center and o_k is the ground-truth offset.

In the evaluation phase, a confidence threshold of 0.3 is applied to valid object detection results.

B. Quantitative Comparison

This section validates the superiority of the proposed method by comparing it with state-of-the-art object detection methods. The comparison includes six traditional methods (GoDec [50], DECOLOR [51], E-LSD [4], B-MCMD [12], D&T [52], MMB [3]) and five deep-learning methods (CenterNet [36],

DSFNet [22], STFNet [53], MICPL [27], MOTIP [54]).

1) Quantitative Comparison Results on Each Test Set: The quantitative comparison results on each test set are shown in Table I, with the best top-three results in each dataset highlighted in red, blue, and green, respectively. From the results, deep learning methods generally outperform traditional approaches. MMB [3] is the strongest traditional method, achieving average Re, Pr, and $F1$-scores of 50.7%, 91.6%, and 64.6%, respectively, because its cumulative frame-difference and matrix-completion strategy partially compensates for weak appearance features. Nevertheless, traditional methods perform

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TABLE I

QUANTITATIVE RESULTS OF VARIOUS METHODS ON TEST SETS; THE TOP-THREE RESULTS ARE SHOWN IN RED, BLUE, AND GREEN, RESPECTIVELY

Method	video 002			video 003			video 005			video 006		
	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑
GoDec [50]	70.5%	79.5%	74.8%	50.5%	86.1%	63.7%	37.7%	87.7%	52.7%	51.9%	81.3%	63.4%
DECOLOR [51]	29.4%	**99.7%**	45.5%	52.8%	90.4%	66.6%	45.9%	86.9%	60.1%	34.2%	98.9%	50.9%
E-LSD [4]	58.8%	77.4%	66.8%	40.5%	41.4%	40.9%	36.2%	92.4%	52.0%	46.4%	90.2%	61.3%
B-MCMD [12]	62.8%	94.4%	75.5%	46.2%	88.0%	60.6%	40.2%	83.0%	54.2%	56.4%	72.2%	63.3%
D&T [52]	59.3%	93.1%	72.4%	43.0%	89.8%	58.2%	40.1%	87.3%	54.9%	59.4%	83.2%	69.3%
MMB [3]	66.5%	**95.9%**	**78.6%**	45.3%	**94.6%**	61.2%	40.9%	**94.7%**	57.2%	63.0%	86.8%	73.0%
CenterNet [36]	60.5%	77.4%	67.9%	56.5%	88.7%	69.0%	47.3%	85.0%	60.8%	69.9%	75.1%	72.4%
DSFNet [22]	72.9%	79.9%	76.2%	**60.8%**	94.3%	**73.9%**	**53.0%**	91.0%	**67.0%**	72.3%	**95.5%**	**82.3%**
STFNet [53]	**77.7%**	93.0%	**84.6%**	58.1%	**94.8%**	72.1%	43.3%	**95.0%**	59.5%	**73.0%**	**95.6%**	**82.8%**
MICPL [27]	**79.9%**	73.0%	76.3%	**63.2%**	93.2%	**75.3%**	**57.2%**	90.1%	**70.0%**	**78.0%**	82.3%	80.1%
MOTIP [54]	32.8%	71.5%	44.9%	43.2%	84.3%	57.1%	38.9%	71.7%	50.5%	38.1%	60.0%	46.6%
CSTANet (Ours)	**78.2%**	**98.3%**	**87.1%**	**71.5%**	**97.4%**	**82.4%**	**63.1%**	**97.0%**	**76.4%**	**80.1%**	**97.5%**	**87.9%**

Method	video 008			video 009			video 010			Average		
	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑
GoDec [50]	45.9%	76.5%	57.4%	45.4%	74.8%	56.5%	24.4%	68.2%	35.9%	46.6%	79.2%	57.8%
DECOLOR [51]	51.5%	88.8%	65.2%	55.5%	77.5%	64.7%	21.1%	89.1%	34.1%	41.5%	90.2%	55.3%
E-LSD [4]	35.6%	87.3%	50.6%	34.0%	81.8%	48.1%	37.8%	74.8%	50.2%	41.3%	77.9%	52.8%
B-MCMD [12]	31.4%	81.6%	45.4%	42.7%	74.1%	54.2%	58.9%	62.2%	60.5%	48.4%	79.4%	59.1%
D&T [52]	36.7%	85.8%	51.4%	41.1%	81.2%	54.6%	**60.9%**	56.9%	58.8%	48.6%	82.5%	60.0%
MMB [3]	40.1%	90.3%	55.5%	42.3%	91.4%	57.9%	**56.5%**	87.5%	**68.6%**	50.7%	**91.6%**	64.6%
CenterNet [36]	59.2%	93.6%	72.5%	53.6%	90.1%	67.2%	44.2%	83.9%	57.9%	55.9%	84.8%	66.8%
DSFNet [22]	**67.4%**	**96.3%**	**79.3%**	**55.5%**	**95.7%**	**70.3%**	46.7%	77.0%	58.1%	**61.2%**	90.0%	**72.5%**
STFNet [53]	55.7%	**96.7%**	70.7%	50.7%	**97.6%**	66.8%	46.8%	**99.9%**	63.7%	57.9%	**96.1%**	71.4%
MICPL [27]	**62.3%**	92.5%	**74.4%**	**55.7%**	93.1%	**69.7%**	51.2%	**99.1%**	**67.5%**	**63.9%**	89.0%	**73.3%**
MOTIP [54]	31.7%	92.7%	47.3%	36.3%	81.7%	50.3%	40.7%	59.3%	48.3%	37.4%	74.5%	49.3%
CSTANet (Ours)	**69.2%**	**98.7%**	**81.4%**	**63.4%**	**96.9%**	**76.7%**	**53.7%**	**98.2%**	**69.5%**	**68.5%**	**97.7%**	**80.2%**

The dashed line separates the traditional-based and deep learning-based algorithms.

competitively only on isolated sequences and lack consistent generalization.

Among deep models, CenterNet [36] achieves only 66.8% $F1$, merely 2.4 percentage points above MMB [3], showing the limitation of relying solely on single-frame spatial features. MOTIP [54] performs worst, with 37.4% Re, 74.5% Pr, and 49.3% $F1$. Its query-based deformable DETR detector is less suited to weak, low-contrast targets because small-object responses can be suppressed by complex backgrounds without task-specific temporal enhancement. On the contrary, DSFNet [22], STFNet [53], and MICPL [27] perform better by incorporating temporal or spatio-temporal information, with MICPL obtaining the strongest baseline $F1$ of 73.3%.

The proposed CSTANet achieves the highest $F1$-score on every sequence and average Re, Pr, and $F1$-scores of 68.5%, 97.7%, and 80.2%, respectively. Compared with the second-best MICPL [27], these values improve by 4.6, 8.7, and 6.9 percentage points. Moreover, while DSFNet [22] and MICPL [27] achieve slightly higher recall than STFNet [53], their precision is lower. On the contrary, CSTANet

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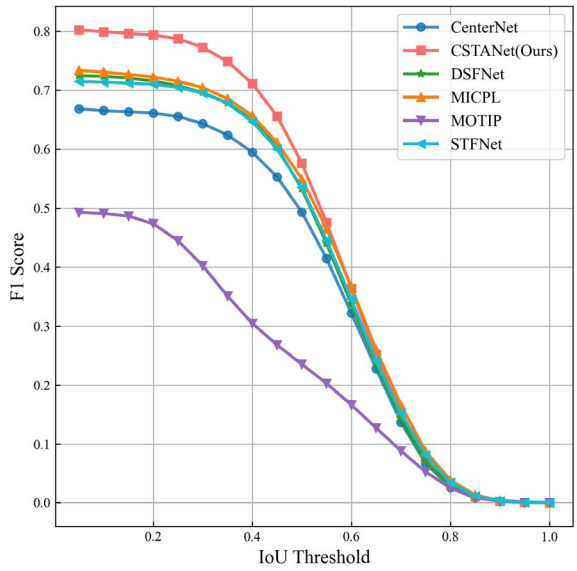


Fig. 7. $F1$-score curves of the different comparison methods under different IoU thresholds.

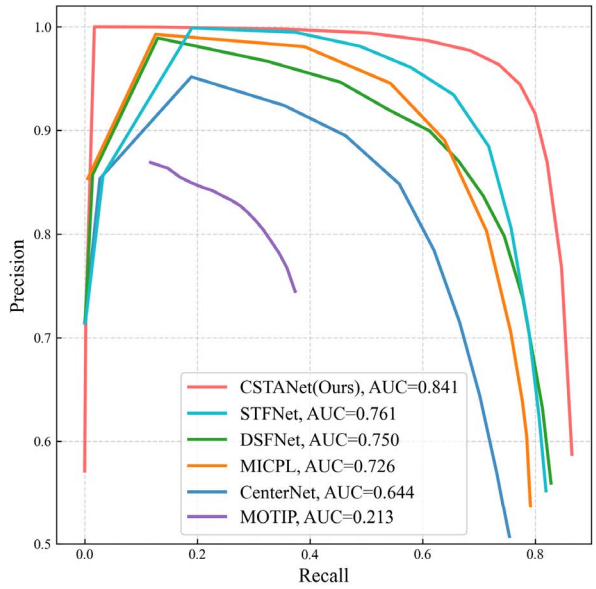


Fig. 8. P–R curves of the different comparison methods under different confidence thresholds.

significantly improves both recall and precision. This indicates that CSTANet effectively balances recall and precision when detecting small moving objects in satellite remote sensing videos, thereby achieving higher overall detection performance.

2) Evaluation of the Overall Performance of Different Methods: Fig. 7 compares $F1$-scores under different IoU thresholds. MOTIP [54] remains at a low level across most thresholds, indicating that its query-based detector has difficulty localizing weak small targets. CenterNet [36] also underperforms the temporal models, confirming the importance of effective spatio-temporal modeling. On the contrary, CSTANet consistently remains above the other methods and still achieves 57.6% $F1$ at an IoU threshold of 0.5. This result highlights CSTANet's outstanding robustness and detection performance, enabling it to adapt to detection tasks under various IoU thresholds. This is attributed to CSTANet's sophisticated TFE network and optimized spatio-temporal feature fusion strategy, which allows it to reliably capture temporal cues of objects in satellite videos, thereby significantly enhancing detection effectiveness.

Fig. 8 shows that CSTANet provides the best precision–recall trade-off, with an area under the curve (AUC)-PR of 0.841, compared with 0.761, 0.750, 0.726, 0.644, and 0.213 for STFNet [53], DSFNet [22], MICPL [27], CenterNet [36] and MOTIP [54], respectively. MOTIP's low AUC indicates that increasing recall causes a substantial loss of precision, whereas CSTANet maintains both metrics over a broader confidence range. This stability is particularly important for the highly imbalanced task of satellite-video small-object detection.

3) Comparison of Network Complexity in Different Methods: To demonstrate the network complexity of different models in detail, we selected FLOPs, parameter size, inference time, memory consumption, and performance-cost-ratio (PCR) as evaluation metrics. These results are displayed in Table II. It should be noted that the memory usage calculated here refers to the peak allocated GPU memory. In addition, the formula for calculating PCR is as follows:

$$PCR = F1/(1/FPS) = F1^{*}FPS. \tag{12}$$

TABLE II

QUANTITATIVE RESULTS OF NETWORK COMPLEXITY IN DIFFERENT METHODS. THE BEST RESULTS ARE SHOWN IN RED AND THE SECOND-BEST RESULTS ARE SHOWN IN BLUE

Model	#Param↓	Flops ↓	TC ↓	Memory ↓	PCR ↑
CenterNet [36]	**6.51 M**	**146.11 G**	0.26 s	**1561.65 MB**	2.57
DSFNet [22]	**6.71 M**	**193.32 G**	0.29 s	**1785.65 MB**	2.50
STFNet [53]	6.86 M	212.36 G	0.59 s	2556.23 MB	1.21
MICPL [27]	7.93 M	195.08 G	**0.23 s**	2780.63 MB	**3.19**
MOTIP [54]	40.71 M	450.68 G	**0.09 s**	17187.21 MB	**5.24**
CSTANet	7.92 M	239.17 G	0.31 s	3065.38 MB	2.59

TC stands for time cost.

As shown in Table II, CenterNet [36] is the lightest model, whereas MOTIP [54] has the fastest inference but also the largest parameter count, FLOPs, and memory usage while producing the lowest detection accuracy. Our proposed CSTANet has 7.92M parameters, 239.17G FLOPs, 3065.38 MB memory, and 0.31 s per inference. Its computational cost is higher than that of lightweight baselines because of CTDL and MCU, which remains a limitation for real-time deployment. Nevertheless, CSTANet achieves the highest $F1$-score and a PCR of 2.59. As illustrated in Fig. 9, it provides a favorable accuracy–efficiency tradeoff: its additional computation yields substantially higher detection accuracy, while MOTIP's faster inference does not compensate for its poor detection performance and high model complexity.

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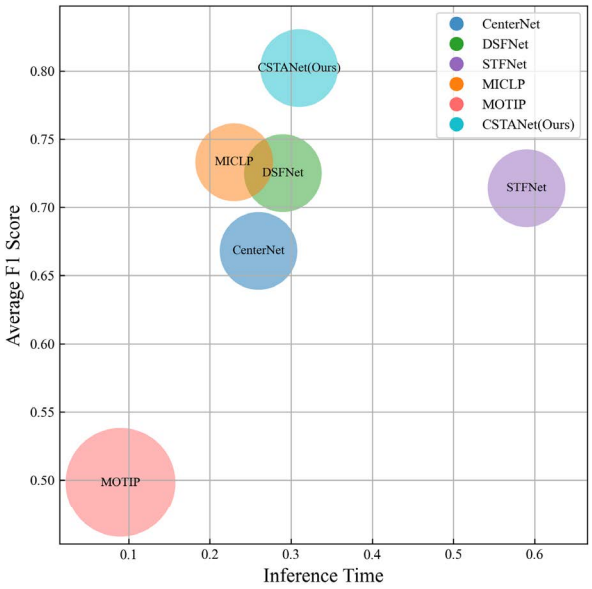


Fig. 9. Bubble chart comparing $F1$-scores, inference times, and parameter sizes of different models. The bigger the bubble size is, the higher the parameter size of the model is.

C. Qualitative Analysis

To more intuitively observe the detection results of different models and evaluate their performance, the detection results of six deep learning models on multiple test sets are visualized, as shown in Fig. 10. Rows in the figure display the qualitative detection results for different test sets, arranged sequentially from top to bottom as video 002, video 003, video 005, video 006, video 008, video 009, and video 010. Each column corresponds to the detection results of a different comparison method. The first column represents the GT for object detection, while columns 2 through 7 represent the detection results of CenterNet [36], DSFNet [22], STFNet [53], MICPL [27], MOTIP [54], and CSTANet, respectively. In each image, green boxes denote TPs, red boxes indicate FPs, and blue boxes represent FNs.

From Fig. 10, it is evident that in densely populated vehicle scenes, some degree of missed detection occurs regardless of the method used, such as in video 002, video 005, and video 010. However, compared with other models, the proposed CSTANet significantly reduces the number of missed detections. For instance, in video 010, CSTANet misses only three objects, far fewer than other models. Furthermore, CSTANet exhibits almost no FPs, with no FPs appearing in any of its detection results.

On the contrary, CenterNet [36], which only utilizes spatial features of objects, shows a higher number of FPs. Although DSFNet [22], STFNet [53], and MICPL [27] reduce the occurrence of FPs to some extent, some FPs are still present. MOTIP [54] shows more obvious detection errors, with a significant increase in FPs and more missed detections than the other comparison methods. This indicates that its detection results are less stable in complex satellite video backgrounds, where weak small-object features are easily confused with background responses or suppressed during detection. Overall, the proposed CSTANet demonstrates the best detection performance across all test video sets while exhibiting exceptional robustness, effectively handling complex scenarios. By deeply exploring cross-frame temporal information and fully integrating spatio-temporal data, CSTANet significantly enhances the detection performance of small moving objects in satellite videos.

D. Robustness Analysis

To further verify the robustness of the proposed method, we conducted additional comparative experiments on the SatMTB-MOT [30] dataset. From the perspective of temporal modeling, the compared methods include cross-frame difference/LSTM-based CKDNet [56], 3-D or sparse spatio-temporal convolution-based DSFNet [22] and Hieum [23], recurrent or memory-based MP^{2}Net [26] and SPAX [25], and cross-frame feature updating-based CFTrack [55]. Although these methods adopt different temporal modeling strategies, some of them also introduce attention-related designs in other modules. In particular, CFTrack [55] uses attention-inspired spatial/channel enhancement, MP^{2}Net [26] adopts patch-wise cross-attention for local feature enhancement, and SPAX [25] employs a multidimensional attention block in its asymmetric feature interaction module.

As shown in Table III, CSTANet achieves the strongest overall robustness across vehicle, airplane, and ship categories. For vehicles, CSTANet obtains the highest Re and $F1$-scores of 74.9% and 81.3%, respectively. Compared with Hieum, which uses long-term sparse spatio-temporal modeling, CSTANet improves $F1$ by 4.0%. Compared with attention-involved CFTrack [55], MP^{2}Ne [26], and SPAX [25], CSTANet improves $F1$ by 10.1%, 9.8%, and 5.7%, respectively. Although DSFNet [22] and MP^{2}Net [26] achieve higher precision, their recall scores are only 59.6% and 57.1%, indicating that they still miss many weak or densely distributed vehicle targets. For airplanes, CSTANet reaches 86.9% Re and 89.9% $F1$, outperforming SPAX [25] by 4.6% and 2.5%, respectively, while maintaining a comparable precision of 93.0%. This shows that CSTANet can better preserve target responses under appearance ambiguity. For ships, CSTANet achieves the highest recall of 70.8%, exceeding SPAX [25] and MP^{2}Net [26] by 8.0% and 10.5%, respectively. Its $F1$-score is slightly lower than SPAX [25] due to lower precision, suggesting that CSTANet tends to recover more difficult ship targets while introducing limited additional FPs. On average, CSTANet achieves 77.5% Re and 80.5% $F1$, surpassing the strongest baseline SPAX [25] by 7.8% and 2.1%. These results demonstrate that CSTANet generalizes well across different object categories, motion patterns, and background conditions, showing stronger robustness than existing temporal modeling strategies.

E. Ablation Study

To investigate the effectiveness of different modules and hyperparameter settings in this model, we design seven groups of ablation experiments. The test subsets used for these ablation experiments are consistent with the test sets, as shown in

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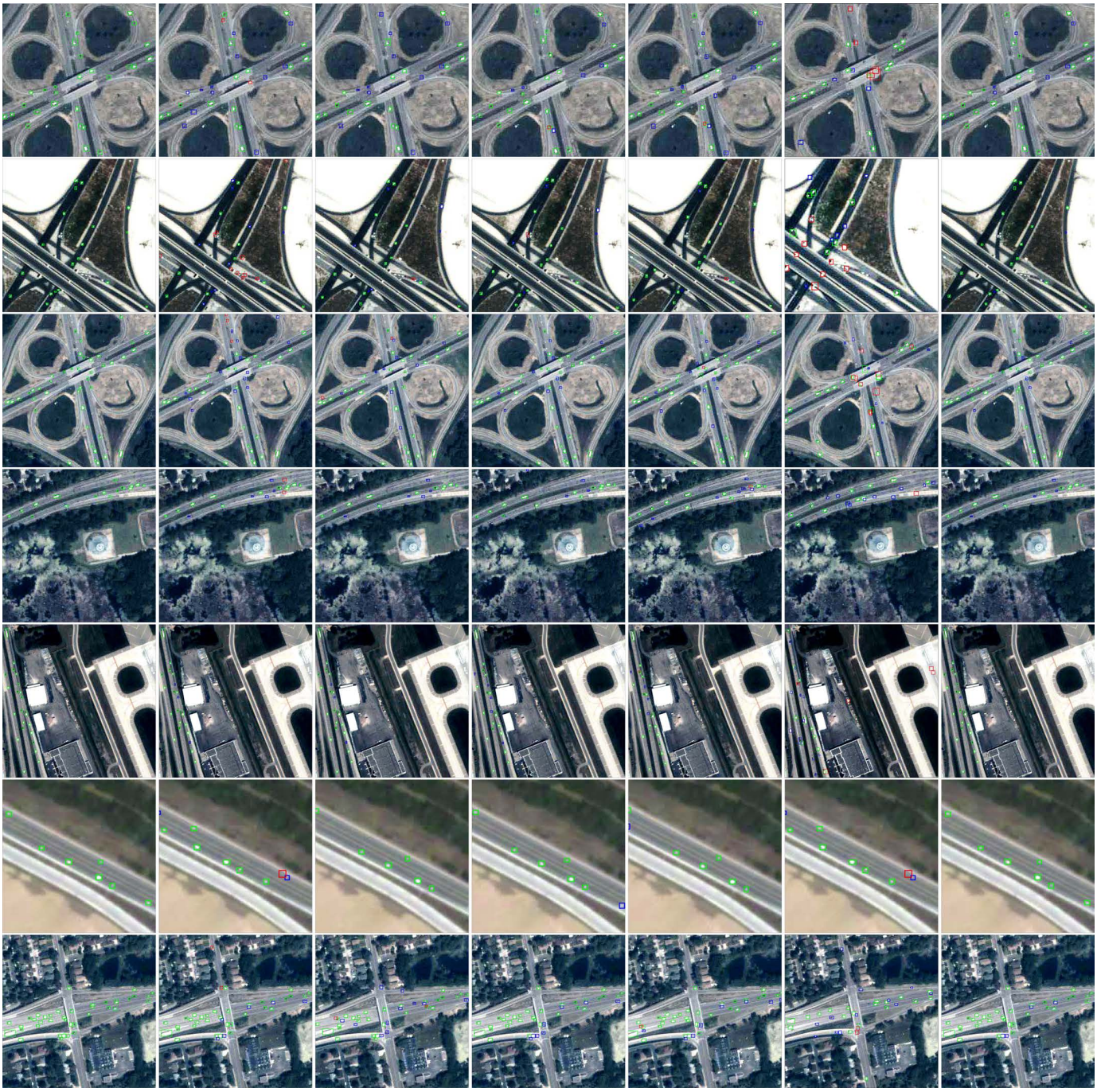


Fig. 10. Qualitative results of different comparison methods on the test dataset. Each row presents results from a distinct test set, arranged in the order of videos 002, 003, 005, 006, 008, 009, and 010. Each column corresponds to the detection results of comparison methods, where green boxes denote TPs, red boxes indicate FPs, and blue boxes represent FNs. Column labels are (a) GT, (b) CenterNet [36], (c) DSFNet [22], (d) STFNet [53], (e) MICPL [27], (f) MOTIP [54], and (g) Ours (CSTANet).

Fig. 6. For generality, the average value of each metric across the seven subsets is used as the evaluation criterion.

The experiments evaluate module combinations, TFE and MCU computational strategies, the two-stage CTDL compensation, MCU depth, input-frame number, and frame interval. CenterNet [36] is used as the baseline. Tables IV–VII present the detection results of models with different components. To better visualize the detection performance of models with different components, detection results for test set 005 are illustrated in Fig. 11. Furthermore, Table VIII presents the detection results under different depths of residual blocks (DoRBs) in the MCU, analyzing the impact of this crucial architectural parameter on the quality of spatio-temporal feature alignment and overall detection performance. Finally, Tables IX and X show the detection results under different input frame counts and frame intervals, analyzing

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TABLE III

QUANTITATIVE RESULTS OF VARIOUS METHODS ON SATMTB-MOT, THE TOP-THREE RESULTS ARE SHOWN IN RED, BLUE, AND GREEN, RESPECTIVELY

Method	Vehicle			Airplane			Ship		
	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑	Re ↑	Pr ↑	F1 ↑
CFTrack [55]	63.4%	81.1%	71.2%	70.7%	72.7%	71.7%	**63.9%**	65.1%	64.5%
CKDNet [56]	37.9%	76.2%	50.6%	52.6%	**96.8%**	68.1%	29.5%	70.8%	41.6%
DSFNet [22]	59.6%	**94.6%**	73.1%	66.9%	**95.1%**	78.5%	53.6%	**82.2%**	64.9%
Hieum [23]	**74.0%**	81.0%	**77.3%**	59.3%	92.0%	72.1%	37.0%	74.7%	49.5%
MP^{2}Net [26]	57.1%	**95.6%**	71.5%	**75.9%**	**96.1%**	**84.8%**	60.3%	**79.6%**	**68.6%**
SPAX [25]	**64.2%**	**91.9%**	**75.6%**	**82.3%**	93.1%	**87.4%**	**62.8%**	**85.6%**	**72.4%**
CSTANet (Ours)	**74.9%**	89.0%	**81.3%**	**86.9%**	93.0%	**89.9%**	**70.8%**	77.5%	**70.4%**

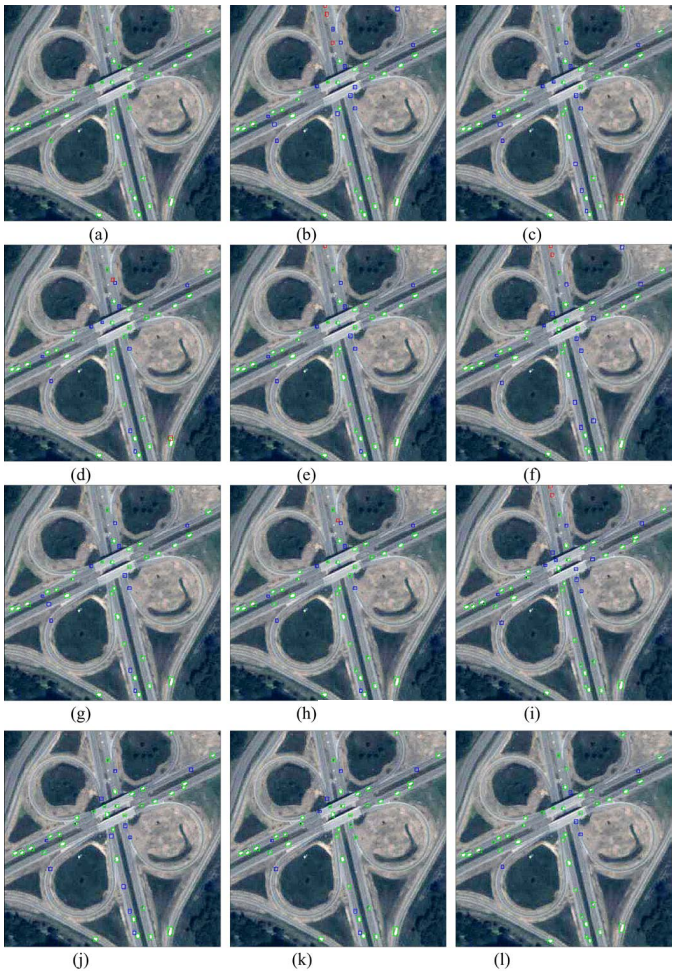


Fig. 11. Qualitative detection results of different models on test video 005. Green boxes represent TPs, red boxes indicate FPs, and blue boxes denote FNs. (a) GT. (b) Model 1. (c) Model 2. (d) Model 3. (e) Model 4. (f) Model 5. (g) Model 6. (h) Model 7. (i) Model 8. (j) Model 9. (k) Model 10. (l) CSTANet (Ours).

the impact of these two hyperparameters on detection performance.

TABLE IV

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON MODULE EFFECTIVENESS

Model	Baseline	TFE	MCU	Re ↑	Pr ↑	F1 ↑
Model 1	✓	-	-	55.9%	84.8%	66.8%
Model 2	-	✓	-	62.5%	94.2%	74.6%
Model 3	-	✓	✓	68.4%	91.0%	77.5%
Model 4	✓	✓	-	67.1%	96.1%	78.6%
CSTANet	✓	✓	✓	**68.5%**	**97.7%**	**80.2%**

For each column, the optimal result is highlighted in bold. TFE stands for temporal feature extraction, and MCU refers to the motion compensation unit.

TABLE V

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON TFE COMPUTATIONAL STRATEGY

Model	Concat	Diff	Re ↑	Pr ↑	F1 ↑
Model 5	✓	-	57.4%	88.3%	69.3%
CSTANet (Ours)	-	✓	**68.5%**	**97.7%**	**80.2%**

For each column, the optimal result is highlighted in bold. Concat and Diff denote the feature concatenation and difference operations, respectively.

TABLE VI

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON MCU COMPUTATIONAL STRATEGY

Model	Sum	CA	ResBlock	Re ↑	Pr ↑	F1 ↑
Model 6	✓	-	-	63.2%	97.7%	75.8%
Model 7	-	✓	-	65.2%	93.6%	76.3%
CSTANet (Ours)	-	-	✓	**68.5%**	**97.7%**	**80.2%**

For each column, the optimal result is highlighted in bold. Sum and CA denote the feature summation and channel attention operations, respectively.

1) Effectiveness of Different Modules: Table IV evaluates different combinations of the baseline, TFE, and MCU. The corresponding quantitative detection results are illustrated in Fig. 11(b)–(e) and (l). CenterNet [36] alone achieves only

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TABLE VII
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON TWO-STAGE COMPENSATION STRATEGY IN THE CTDL

Model	CS 1	CS 2	Re ↑	Pr ↑	F1 ↑
Model 8	-	-	52.8%	91.9%	66.6%
Model 9	✓	-	60.3%	94.6%	72.9%
Model 10	-	✓	62.8%	97.3%	75.8%
CSTANet (Ours)	✓	✓	**68.5%**	**97.7%**	**80.2%**

For each column, the optimal result is highlighted in bold. CS 1 and CS 2 stand for the first compensation stage and the second compensation stage, respectively.

TABLE VIII
QUANTITATIVE RESULTS OF COMPARATIVE STUDY ON DORBS IN THE MCU

DoRB	Re ↑	Pr ↑	F1 ↑
3	60.0%	95.8%	73.3%
4	61.0%	97.4%	74.5%
6	65.6%	96.8%	77.7%
7	56.6%	**97.9%**	71.4%
5 (CSTANet)	**68.5%**	97.7%	**80.2%**

For each column, the optimal result is highlighted in bold.

TABLE IX
QUANTITATIVE RESULTS OF COMPARATIVE STUDY ON NOIFS

NoIF	Re ↑	Pr ↑	F1 ↑
1	55.9%	84.8%	66.8%
3	63.5%	89.1%	74.0%
7	58.2%	**97.7%**	72.5%
9	65.8%	96.5%	77.9%
5 (CSTANet)	**68.5%**	97.7%	**80.2%**

TABLE X
QUANTITATIVE RESULTS OF COMPARATIVE STUDY ON NUMBER OF FRAME INTERVAL (NOFI)

NoFI	Re ↑	Pr ↑	F1 ↑
1	**70.2%**	93.2%	79.7%
2	67.0%	96.7%	78.7%
4	63.8%	**98.0%**	76.8%
5	68.2%	97.4%	79.9%
3 (CSTANet)	68.5%	97.7%	**80.2%**

66.8% $F1$ and produces substantial FPs and missed detections, confirming that spatial appearance is insufficient for moving small objects. TFE alone raises $F1$ to 74.6%, showing the value of temporal cues. Adding MCU further improves recall from 62.5% to 68.4%, although precision decreases to 91.0%, indicating that compensation recovers more targets at the cost of some false alarms.

Combining CenterNet and TFE achieves 78.6% $F1$, while the complete CSTANet reaches the best Re, Pr, and $F1$-scores of 68.5%, 97.7%, and 80.2%. Fig. 11 shows that the full model reduces both FPs and missed detections. These results verify that spatial features, temporal differences, and motion compensation are complementary rather than independently sufficient.

2) Effectiveness of TFE Computational Strategy: The TFE module is one of the key components of this model. This module primarily utilizes temporal differences to extract the temporal cues of the object, thereby improving detection performance. Its computational strategy has a significant impact on detection effectiveness. In this experiment, two different TFE computational strategies are designed: feature concatenation (Concat) and temporal difference (Diff). The experimental results are shown in Table V. The corresponding quantitative detection results are illustrated in Fig. 11(f) and (l).

From Table V, concatenation obtains only 57.4% Re, 88.3% Pr, and 69.3% $F1$ and produces numerous FPs and missed detections. Temporal difference increases $F1$ to 80.2% by explicitly emphasizing inter-frame motion. Therefore, direct concatenation introduces temporal context but cannot isolate weak motion cues as effectively as frame-difference modeling.

3) Effectiveness of MCU Computational Strategies: The primary function of the MCU is to align temporal and spatial features, avoiding spatio-temporal mismatches. Its computational strategy plays a crucial role in detection performance. In this experiment, three distinct MCU computational strategies are designed: feature summation (Sum), channel attention (CA), and residual block (ResBlock). The experimental results are shown in Table VI. The corresponding quantitative detection results are illustrated in Fig. 11(g), (h), and (l).

From Table VI, summation achieves high precision of 97.7% but only 63.2% recall, indicating more missed detections. CA raises recall to 65.2% but reduces precision to 93.6%, increasing false alarms. The residual-block MCU achieves the best Re, Pr, and $F1$-scores of 68.5%, 97.7%, and 80.2%, respectively. This result indicates that simple summation and channel reweighting cannot compensate for spatio-temporal discrepancy as effectively as residual feature correction.

4) Effectiveness of the Two-Stage Compensation Strategy in the CTDL: To evaluate the impact of the two-stage compensation strategy in CTDL, we conduct ablation experiments by selectively enabling each stage. The quantitative results are presented in Table VII, and qualitative detection outcomes are illustrated in Fig. 11(i)–(l).

As shown in Table VII, disabling both stages reduces $F1$ to 66.6% and causes substantial FPs and missed detections. Using only the first stage improves $F1$ to 72.9% by providing coarse temporal alignment, but fine structural mismatches remain. Using only the second stage reaches 75.8% $F1$ and better preserves local boundaries but lacks sufficient global temporal constraints. Activating both stages achieves 68.5% Re, 97.7% Pr, and 80.2% $F1$. Thus, coarse alignment and fine refinement are complementary: the former improves global consistency, whereas the latter corrects local residual errors.

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5) Comparative Study on DoRBs in the MCU: The DoRBs in the MCU are a critical architectural parameter that directly affects the quality of spatio-temporal feature alignment and overall detection performance. To evaluate this, we conducted a comparative study by varying the number of residual blocks in the MCU from 3 to 7. The quantitative results are presented in Table VIII.

As shown in Tables V and VIII residual blocks provide the best performance, with 68.5% Re, 97.7% Pr, and 80.2% $F1$. Shallower configurations lack sufficient capacity to correct motion-induced misalignment, resulting in recalls of only 60.0% and 61.0% for 3 and 4 blocks. Increasing the depth to 6 or 7 blocks also reduces $F1$ to 77.7% and 71.4%, respectively. Excessive spatial refinement can suppress temporal motion cues and create an imbalance between spatial and temporal information. Five blocks therefore provide the best balance between alignment capacity and motion preservation.

6) Comparative Study on NoIFs: The number of input frames (NoIFs) is a crucial hyperparameter affecting detection performance. In this experiment, comparative studies are conducted with different numbers of input frames, specifically using 1, 3, 5, 7, and 9 frames as inputs. The experimental results are shown in Table IX.

The results indicate that five input frames achieve the best $F1$-score of 80.2%. A single frame cannot provide temporal cues and yields only 66.8% $F1$, while three frames provide insufficient temporal context. Increasing the input to seven or nine frames also reduces $F1$ to 72.5% and 77.9%, respectively, because additional frames introduce background noise and computational overhead. At the dataset frame rate of ten frames/s [3], five frames provide sufficient motion context without an excessive temporal window, making this setting more suitable for practical deployment.

7) Comparative Study on Frame Interval: The frame interval is another critical hyperparameter affecting detection performance. This experiment conducts comparative studies using different frame intervals, specifically testing intervals of 1, 2, 3, 4, and 5. The experimental results are shown in Table X.

The results indicate that an interval of three achieves the best offline performance with 80.2% $F1$. Smaller intervals of one and two yield 79.7% and 78.7% $F1$ because adjacent frames contain less distinguishable displacement. Larger intervals of four and five reduce $F1$ to 76.8% and 79.9%, respectively, as increased temporal separation can break motion continuity and lose relevant cues. Thus, interval three provides the best balance for the current benchmark. For practical deployment at ten FPS [3], however, an interval of three may produce discontinuous observations of small or slow-moving targets. Although interval one gives slightly lower offline performance, it preserves denser motion continuity and may therefore be preferable when detection reliability is more important than the benchmark optimum.

F. Limitation and Discussion

Through the extensive experiments and analyses presented above, it is evident that the proposed CSTANet achieves strong performance in the task of detecting small moving objects in satellite videos. Nevertheless, the method still has certain limitations. First, as shown in Table I, although the recall rate of CSTANet is significantly higher than that of other methods, the absolute value remains relatively low, indicating that a considerable number of objects are still missed. By examining the detection results across different test sets, we found that most of the missed detections correspond to relatively dark small targets. Due to their tiny size and the high similarity of their color to the background, these objects are difficult to distinguish, leading to frequent missed detections.

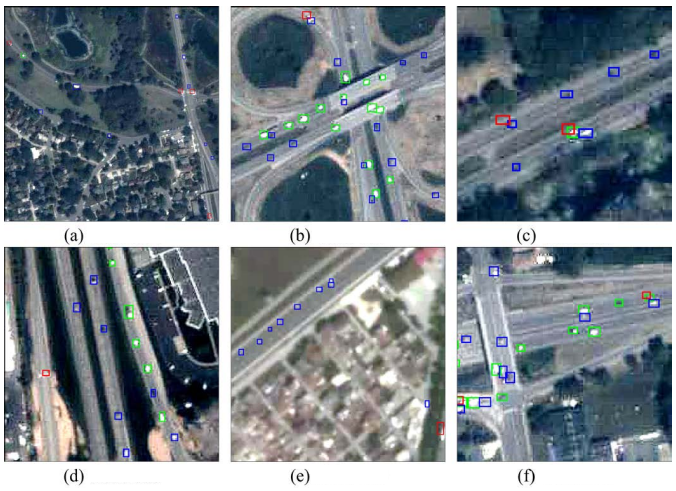


Fig. 12. Failure case of detecting small dark targets in test videos. Green boxes represent TPs, red boxes indicate FPs, and blue boxes denote FNs. (a) Video 002. (b) Video 005. (c) Video 006. (d) Video 008. (e) Video 009. (f) Video 010.

Fig. 12 illustrates several failure cases of such dark small targets in the test set. In future work, we aim to improve the detection rate of dark targets and thereby enhance the overall recall. For example, contrast-adaptive strategies can be applied before feature extraction to enhance the brightness channel and highlight details in dark regions. In addition, temporal attention mechanisms can be employed to fuse multi-frame features, making weak textures clearer through temporal accumulation.

Moreover, as indicated in Table II and Fig. 9, the proposed method involves a relatively large number of parameters, which results in a higher computational cost. Consequently, the inference speed is relatively slow, with an average of about three FPS, which does not meet the real-time detection requirement of 5–10 FPS for satellite videos. In future work, we plan to explore lightweight model architectures and acceleration techniques to achieve real-time detection capability. For instance, unsupervised frameworks such as Hieum [23] could be adopted to significantly improve inference speed.

V. Conclusion

To address the unique challenges of detecting small moving objects in satellite videos, this study proposes a CSTANet. CSTANet employs a CTDL module to extract discriminative temporal cues and an MCU to reduce feature misalignment before spatio-temporal fusion. An anchor-free detection head is further adopted to improve detection efficiency and stability.

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Extensive experiments on the SVMOD and SatMTB-MOT datasets demonstrate that CSTANet outperforms state-of-the-art methods. In particular, experiments on vehicles, airplanes, and ships using SatMTB-MOT show strong overall performance and verify the robustness and generalization capability of CSTANet across different object categories, motion patterns, and background conditions.

Nevertheless, CSTANet still has certain limitations. The overall recall remains relatively low, and the model struggles to detect dark and small targets because of their extremely low contrast with the background. In addition, the network incurs relatively high computational costs and slow inference speed, limiting its applicability in real-time scenarios. In future work, we aim to improve dark-target detection through contrast-adaptive preprocessing and temporal attention-based multiframe fusion. We will also explore lightweight architectures and unsupervised frameworks to enhance recall, reduce computational cost, and accelerate inference for practical satellite video applications.

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